L03 Reflective Journal_FA26_ML_6171_163474

Diane Lenhoff

1. Which type of machine learning interests you most and why?
Supervised learning interests me the most because I like being able to train a model with data where we already know the answers and then see how well it can predict new data. The wine classification example helped me understand how this works.

2. What was the most challenging part of the ML workflow for you?
The most challenging part for me was understanding why the different models gave different results even though they were using the same data. Comparing the models helped me understand this better.

3. How might you apply these concepts to a problem in your field of interest?Health risI think machine learning could be really useful in healthcare. It could look at things like a patient's symptoms, lab results, and medical history to help identify hks or possible diseases.

4. What questions do you have about machine learning that you'd like to explore further?

I would like to understand more about how you decide which model to use for a certain problem. I also want to learn more about how models improve over time and how we know they will work well with new data.

Key Learnings
Most important concept learned: I learned that there are different types of machine learning and different models that can be used depending on the problem. I also learned that preparing and understanding the data is an important part of the process.
Most challenging part: [The most challenging part for me was understanding why different models can give different results even when they are using the same data.
Questions for further exploration: I would like to learn more about how you decide which model to use and how you know if a model will work as well with new real-world data.

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Professor Rao

ITAI 1371: Intro to Machine Learning

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L03 Reflective Journal

Introduction

The purpose of this lab was to understand the machine learning workflow and the differences between the types of learning by working through a notebook on a Wine dataset, from data loading to model evaluation and comparing two models that use different algorithms. This journal focuses on what I learned about why certain models suit certain situations, and why a model must generalize rather than fit the data it was trained on.

What I Learned

Logistic Regression

I initially thought this algorithm involved comparing the features against each other, as if the feature with the highest value decided which class a wine belonged to. I also thought a single strong feature could carry the prediction on its own. It took a while to understand that the comparison isn't between features at all. Each feature value is multiplied by a learned weight, those results are summed for each class, and the class with the highest sum is the prediction.

Memorization vs Generalization

When I found that Logistic Regression worked better than the Decision Tree, I asked myself what would make the Decision Tree outperform the former. I thought changing the max_depth value would help increase the splits, thus more questions, making it a better fit. But I realized I was only tuning it to the specific dataset I was working on, which was leaning towards memorization and overfitting. That’s when it clicked: what makes a model perform well isn’t how closely it fits the data it was trained on, but how well it holds up to **unseen** data. A model that scores 100% strictly on a training set will most likely fail miserably on new data. Too flexible and it leans towards memorizing the training set. Too restricted and it won’t recognize the pattern at all. It must have a balance between the two to maintain overall performance.

Learning Types

Working through the lab made the types of learning more concrete by seeing how some of them worked firsthand. The workflow on the Wine classification used supervised learning, which was made clear by one line: y = df[‘wine_class’], served as the ground truth the samples could check against. Having that already provided makes things much easier than having to produce the ground truth yourself beforehand. The notebook didn’t use an unsupervised model because the labels were already there.

Conclusion

Learning how a model actually works changed my understanding of what makes a model “better.” Thinking that a deeper tree would produce better results only uncovered the overfitting trap that I almost let myself fall into. That’s what allowed me to realize the difference between

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Before I began this module, I had almost no knowledge of the machine learning process or what it looked like. I didn't know there were different styles of learning or how they differ. To admit, I did find it a bit intimidating. Loading up the wine dataset was simple, I could tell that it had 178 entries with 13 features to describe the wine, and 3 classes. There were no missing values, but I could see that all the classes were not equally represented. Class 1 had 71, Class 0 had 59, and Class 2 had 48. I was not sure if that would affect the machine learning process, but it was something I noted for future possible use.

After loading and exploring the data, I went on the plot for visual reference, and I see why this is one of the most important parts of the machine learning workflow. I was especially interested by the correlation map, though I had trouble understanding it initially, I was able to understand it more as I studied it. For example, I noticed that there was a correlation between alcohol content and total phenols, the higher presence in one feature led to a higher presence in the other. Afterwards, I went through all the steps of the ML workflow. First to prep the data, then went on to split it into a training and testing sets. I would like to expand data splitting in the future and why I have to split it the way that we do, but for now I can comprehend that it is to teach the model and test unseen data.

After training and evaluating the data, I found that the logistic regression model had an accuracy of about 89% and the decision tree had about 83%.

Onto model interpretation, I noticed while looking at the confusion matrix that all the values were not on the diagonal, indicating that the model was not perfect. Lastly, I had the chance to build my own data model. After much experimenting with different features, I was able to improve the accuracy with the model features['alcohol', 'total_phenols', 'ash'], leading to an model accuracy of 91.7%. This lab helped me to learn the step-by-step process of training a model through workflow and with hands on experience. It also helped me understand the different styles in which machines learn and how they evaluate data. I plan to study more to differentiate between supervised, unsupervised and reinforcement learning.

KeiChara Fleeks-Jones

My Machine Learning Lab Reflection Journal

Highlight 1: Understanding the End-to-End ML Workflow

One of the biggest things I took away from this lab was a better understanding of the full Machine Learning process. Before completing this lab, I mostly thought of machine learning as training a model and testing how well it worked. I understand there are many crucial steps that need to take place prior to the actual model training then afterwards as well.

First you need to define the problem. Then you have to collect and explore the data, clean and prep it, go on to choose any desired features, select & train a model, evaluate the results, and then eventually you can think about how the model would be deployed and monitored.

A key takeaway was the importance of building the early stages. Taking time to explore the data ensuring accuracy and picking the features can make a huge difference in how the model performs. This lab helped me see machine learning as an ongoing process rather, than just model training.

Highlight 2: The Impact of Feature Selection (Part 7 Hands-On)

One part of the lab that really stood out to me was Part 7, where I tested different combinations of features with the Logistic Regression model. At first, I used the original features and got an accuracy of 88.9%. When I changed the features to malic_acid, magnesium, and OD280/OD315 of diluted wines, the accuracy dropped to 86.1%, showing me that these particular features are not great predictors.

Watching the results change immediately helped me understand how much feature selection can affect a model's performance. Simply adding or changing features does not automatically make a model better, what matters more is choosing features that actually have a meaningful relationship with what the model is trying to predict. This allowed me to see why feature engineering and feature selection are important parts of machine learning.

Personal Learnings and Future Thoughts

This lab helped me see how machine learning works in real life. I understand that building a model is not a set it and forget about it type of thing because it really takes time and effort to test the model and make it work as intended. I would like to learn more about hyperparameter tuning and feature scaling because I want a better understanding of how they actually make things better. The examples we looked at really showed me how powerful these tools can be.

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Going into this lab, I understood supervised, unsupervised, and reinforcement learning as three separate categories, but I hadn't fully thought about what actually separates them in practice. Working through the wine classification project made the distinction clearer, since we had both the chemical measurements and the correct wine class for every sample. That is what made the problem supervised - the answer key was there the whole time, and the model's job was just to learn how the features connected to it.

The part of the workflow that stood out most to me was comparing two different models on the same data. I went in assuming a more complex model like the Decision Tree would automatically perform better than something as simple as Logistic Regression, but the results showed the opposite - Logistic Regression reached 88.9% accuracy while the Decision Tree only reached 83.3%. That surprised me at first, but it made sense once I considered that the Decision Tree was capped at a max depth of 3, which limited how much it could actually learn from the data. It reinforced that a more complicated model isn't always the better choice, and that testing multiple approaches is a real part of the workflow rather than a formality.

I also found the confusion matrix useful for understanding model performance beyond a single accuracy number. Seeing where the model actually made mistakes, rather than just how often, made it clearer why looking at precision and recall separately for each class matters, especially since the wine classes weren't perfectly balanced.

Some of the code cells were harder to follow than others, particularly the model training and evaluation steps, since there was a lot happening in a short amount of code. Working through them slowly and connecting each function call back to the 6-step ML workflow helped it click - data preparation, splitting, training, evaluation, and interpretation are really just the same process every time, applied to different data.

This lab also got me thinking about how this workflow could apply outside of a classroom dataset. I'm interested in how companies in the AI hardware space, like NVIDIA or AMD, might use supervised learning to catch chip defects during manufacturing before they reach customers, using sensor and test data the same way we used chemical measurements to classify wine. Overall, this lab helped me see the ML workflow as a repeatable process rather than something unique to each new problem, and it made me more comfortable with the idea of testing and comparing models instead of assuming one approach is automatically correct.